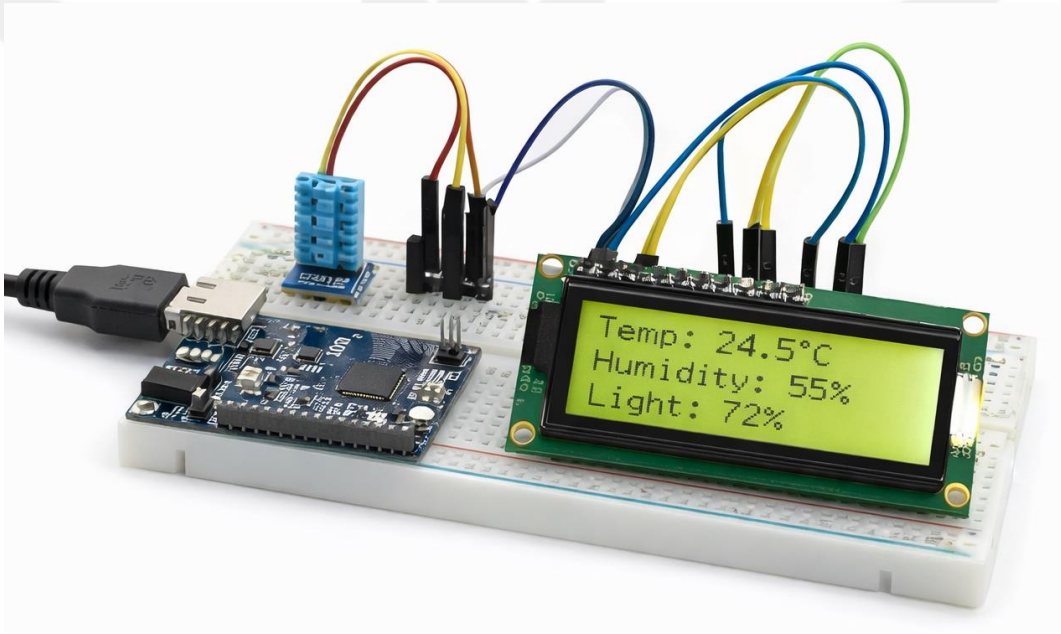
LAUNCH THE PROJECT - 01

Smart Weather Monitoring System

(Intermediate level Project)

On AYNOP® UNO Launchpad Kit



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1. Overview

This project demonstrates how to build a **Smart Weather Monitoring System** using the Arduino® UNO R4 Minima.

The system measures **temperature**, **humidity**, and **ambient light level**, and displays all values in real time on a **16×2 I2C LCD**.

You will learn to:

- Interface a **DHT11 sensor** for temperature and humidity measurement.
- Read analog values from an **LDR (Light Dependent Resistor).**
- Convert raw sensor data into meaningful values.
- Display multiple parameters on a **16×2 I2C LCD**.

This project introduces the concept of **sensor data aggregation and visual display**, making it an ideal intermediate-level application.

2. Components Required

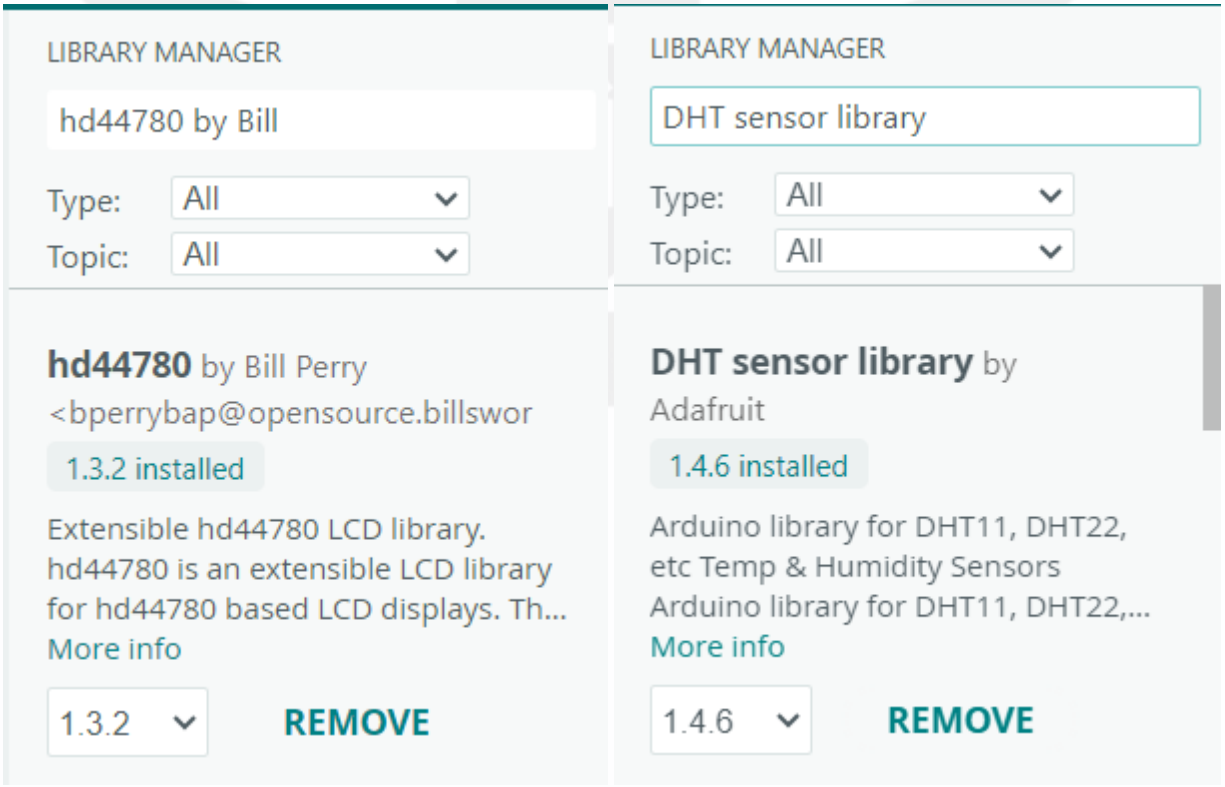
- Arduino® UNO R4 Minima board
- USB Type-C data cable
- 1 × 16x2 LCD with I2C adapter
- DHT11 Temperature & Humidity Sensor
- LDR (Light Dependent Resistor)
- 10 kΩ resistor (for LDR voltage divider)
- Jumper wires
- Breadboard

3. Software Required

- Arduino IDE (v2.3.7 or later)
- Required library - 01: **DHT library**.
- Required library – 02: **hd44780** for 16x2 I2C LCD Display

Installing the Library:

- Open Arduino IDE.
- Go to **Sketch → Include Library → Manage Libraries**
- Search for **DHT sensor library (by Adafruit).**
- Install the library.
- Search for **hd44780 by Bill.**
- Install the library.



Note:
We assume the **Arduino UNO R4 Minima board package** is already installed on your machine, as explained in the 00_Getting_Started/00_GettingStarted_Arduino_R4_Minima guide. If it is not installed, please refer to that document and complete the installation before proceeding.

4. Hardware Setup

This section explains how to connect the sensors and LCD to build the Smart Weather Monitoring System.

4.1 Wiring Diagram

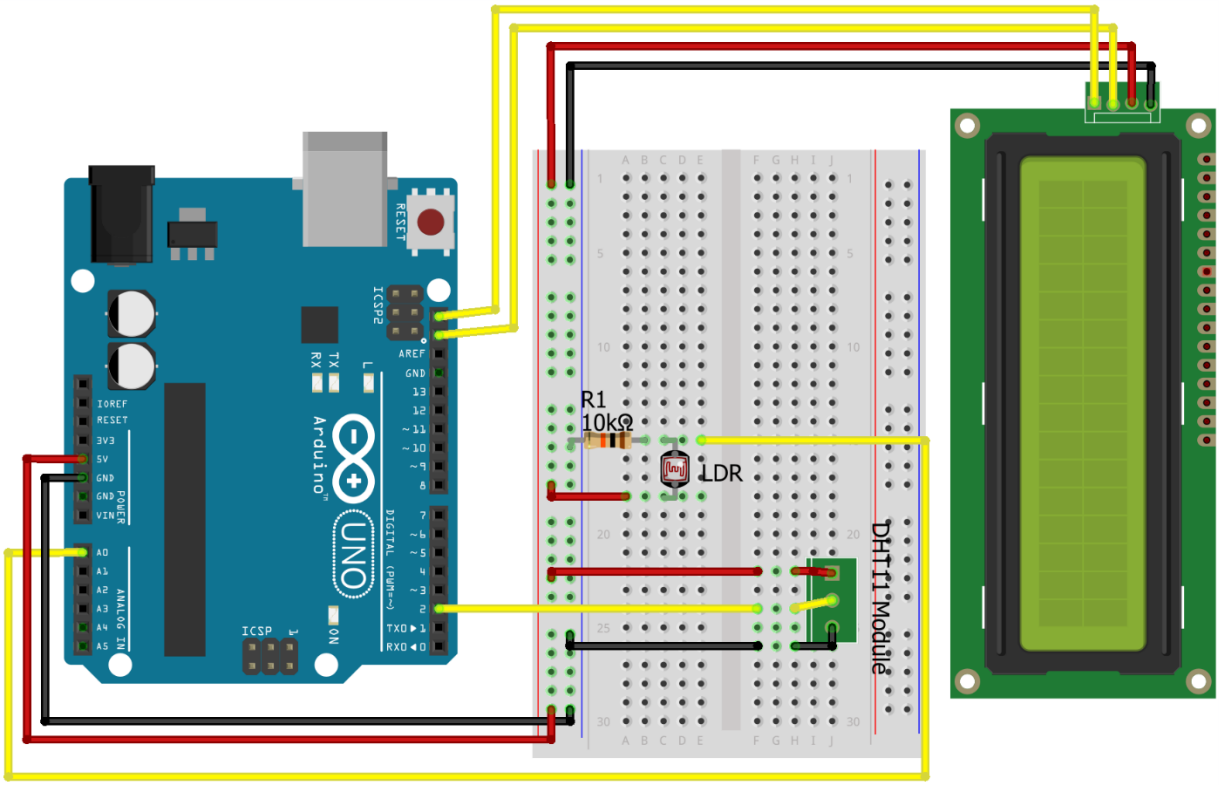


Figure 4.1 – Wiring diagram for Smart Weather Monitoring System

Tip: I2C uses the same two wires (SDA, SCL) for multiple devices. Each device has a unique **I2C address**.

5. Principle – How It Works

This system combines multiple sensors to monitor environmental conditions.

Temperature & Humidity Measurement:

- The DHT11 sensor contains a temperature sensor and a humidity sensor.
- It sends digital data to Arduino over a single data pin.
- The Arduino reads this data using the DHT library and extracts:
    - Temperature in degrees Celsius (°C)
    - Humidity in percentage (%)

Light Measurement:

- The LDR value changes its resistance based on light intensity.
- It forms a voltage divider with a fixed resistor.
- Arduino reads the resulting voltage on analog pin A0.
- The raw analog value is mapped to a light percentage (0–100%).

Display Output:

- The 16×2 I2C LCD displays:
    - Temperature and humidity on the first line.
    - Light level percentage on the second line.
- The display updates every 2 seconds, providing real-time monitoring.

6. Procedure – Steps to Run

1. **Build the Circuit**
   a. Connect the DHT11, LDR, and LCD as shown in the wiring diagram (Figure 4.1).

2. **Connect the Board**
   a. Use a USB Type-C data cable to connect your UNO R4 Minima to your computer.

3. **Open the Project Code**
   a. Simply **double-click** the file *01_Smart_Weather_Monitoring_System.ino* in the project folder, and it will open directly in the Arduino IDE (if installed).

4. **Confirm Board Selection**
   a. The IDE usually auto-detects the UNO R4 Minima if the package is installed.
   b. If not installed, refer to the *00_Getting_Started/00_GettingStarted_Arduino_R4_Minima* document to install the necessary board package.
   c. Verify that *Arduino UNO R4 Minima* is selected in the IDE's board selector (top toolbar).

5. **Upload the Code**
   a. Click the **Upload** button (arrow icon) in the top-left corner of the IDE.
   b. Wait until the console displays "**Done uploading**."

6. **Observe the Behaviour**
   a. The LCD will start displaying temperature, humidity, and light level
   b. Values update automatically every few seconds.

7. Expected Output

- **Light Sensor Test (LDR):**
  - Place your **hand over the LDR** to block light.
  - Observe that the **Light (%) value decreases** on the LCD.
  - Remove your hand, and the **Light (%) value increases** again.

- **Temperature Sensor Test (DHT11):**
  - Bring any **warm object** (for example, your hand or a warm cup) close to the DHT11 sensor.
  - Observe that the **Temperature (°C) value increases** on the LCD.
  - Move the object away and allow the sensor to cool; the temperature value gradually decreases.

8. Code

The source code for this project is included in the downloaded folder:

📂 uno-launchpad-kit/02_Intermediate_Projects/01_Smart_Weather_Monitoring_System/01_Smart_Weather_Monitoring_System.ino

👉 **Tip:**

- To open the project, simply **double-click the .ino file**. If the Arduino IDE is installed, it will launch automatically and load the code.
- If you **haven't installed the Arduino IDE yet**, please refer to:
  📂 uno-launchpad-kit/00_Getting_Started/00_GettingStarted_Arduino_R4_Minima
  to **download and install it.**

8.1 Function References

- setup() – Runs once when the board is powered on or reset. Used to initialize sensors, LCD, and I/O pins.
- loop() – Runs continuously after setup() finishes. Reads sensor values and updates the LCD at fixed intervals.
- dht.begin() – Initializes communication with the DHT11 temperature and humidity sensor.
- dht.readTemperature() – Reads the current temperature value (in °C) from the DHT11 sensor.
- dht.readHumidity() – Reads the current humidity value (in %) from the DHT11 sensor.
- analogRead(pin) – Reads the analog voltage from the LDR circuit connected to the specified analog pin.
- map(value, fromLow, fromHigh, toLow, toHigh) – Converts the raw analog value from the LDR into a percentage range (0–100%).
- lcd.init() – Initializes the 16×2 I2C LCD module.
- lcd.backlight() – Turns ON the LCD backlight for better visibility.
- lcd.clear() – Clears all text currently displayed on the LCD.
- lcd.setCursor(column, row) – Sets the cursor position on the LCD screen.
- lcd.print() – Displays text or numeric values on the LCD.
- delay(ms) – Pauses program execution for a fixed time interval between sensor updates.

📚 **Library Reference:**

📎 Wire Library — I2C communication library used internally by hd44780.
📎 hd44780 — I2C LCD control library used to interface the 16×2 display.
📎 DHT.h — Used to interface with the DHT11 temperature and humidity sensor.

📚 **For more details and advanced usage, visit:**
📎 Arduino Language Reference — The official guide for all Arduino functions.

9. Troubleshooting Tips

- **Error while compiling (hd44780 library not found)?**
  - Open Arduino IDE → Library Manager.
  - Search for "hd44780 (by Bill Perry)" and install it.
  - Restart the IDE and compile again

- **LCD backlight ON but no text?**
  - Adjust the small **blue potentiometer** on the I2C adapter to set contrast.

- **LCD backlight not ON?**
  - I2C adapter boards have a small **jumper near the backlight pins**.
  - Ensure this jumper is connected; otherwise, the backlight will stay OFF

- **LCD shows nothing?**
  - Check wiring: VCC → 5V, GND → GND, SDA → D18, SCL → D19.
  - Verify the **I2C address** (0x27 or 0x3F)

- **Upload error in Arduino IDE?**
  - Verify that the correct board (**Arduino UNO R4 Minima**) is selected in the IDE.
  - Check that the correct **COM port** is chosen.

- **"Sensor Error" message on LCD?**
  - Check DHT11 wiring (VCC, GND, DATA → D2).
  - Ensure the DHT sensor library is installed correctly.
  - Loose connections may cause invalid readings.

- **Temperature or humidity shows incorrect values?**
  - Wait a few seconds after power ON for stable readings.
  - Avoid touching the sensor while testing.
  - Keep away from direct heat or airflow.

- **Light value always 0% or 100%?**
  - Check the LDR voltage divider wiring.
  - Ensure the 10 kΩ resistor is connected between A0 and GND.

- **Still not working?**
  - Recheck all wiring connections carefully.
  - Disconnect power and rebuild the circuit step-by-step.
  - Test each module (LCD, DHT11, LDR) individually.

10. License

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11. Support & Feedback

We value your feedback and are happy to assist with any questions, troubleshooting, or suggestions you may have.

Email: *support@aynop.com*

When sending an email, please include your kit name (AYNOP® UNO Launchpad Kit) and, if applicable, the project name in the subject line. This will help our team respond faster and more accurately. We aim to respond to all queries within 2–3 business days. Your feedback helps us improve our products and create even better learning experiences for all Arduino beginners.